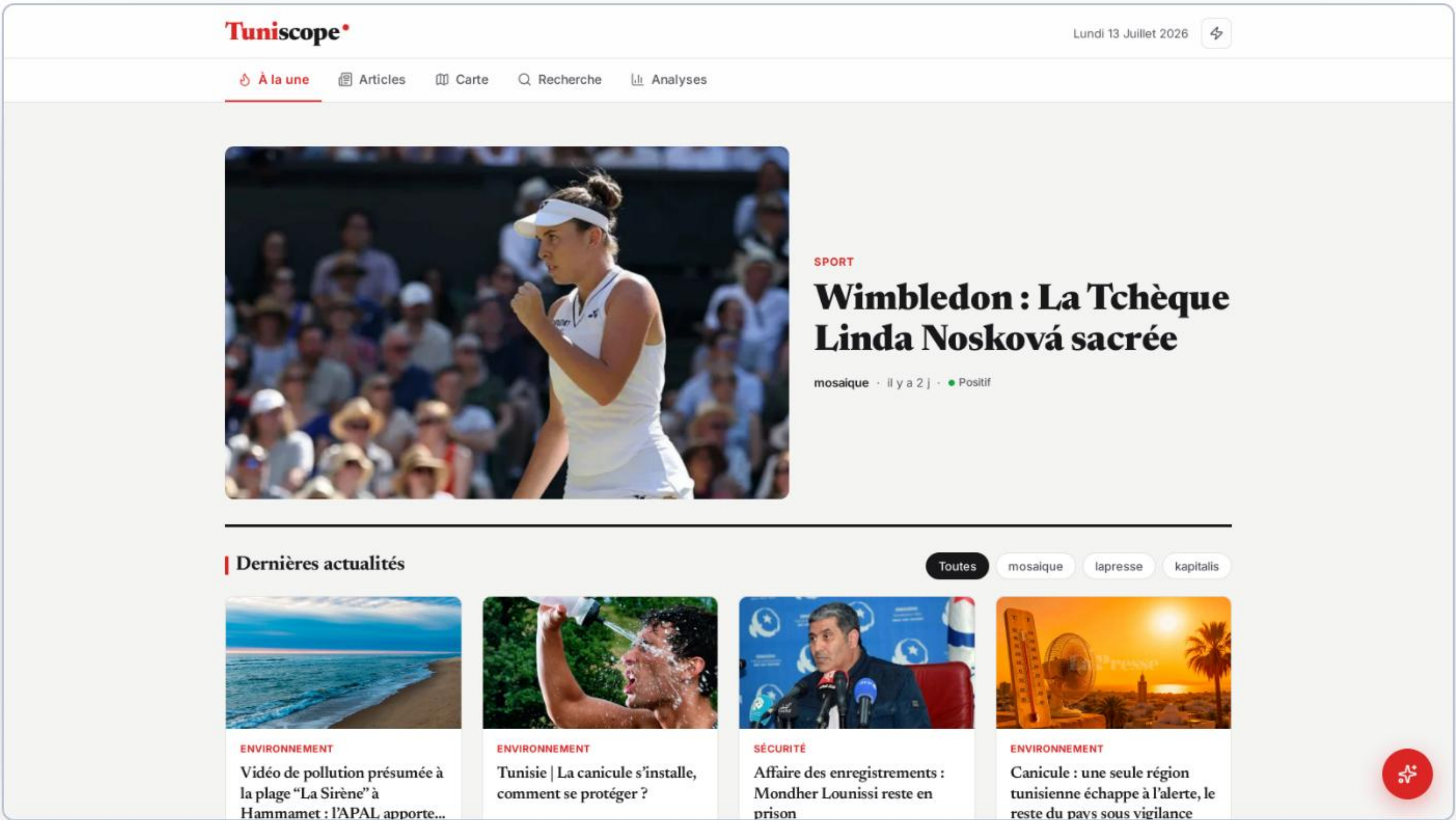
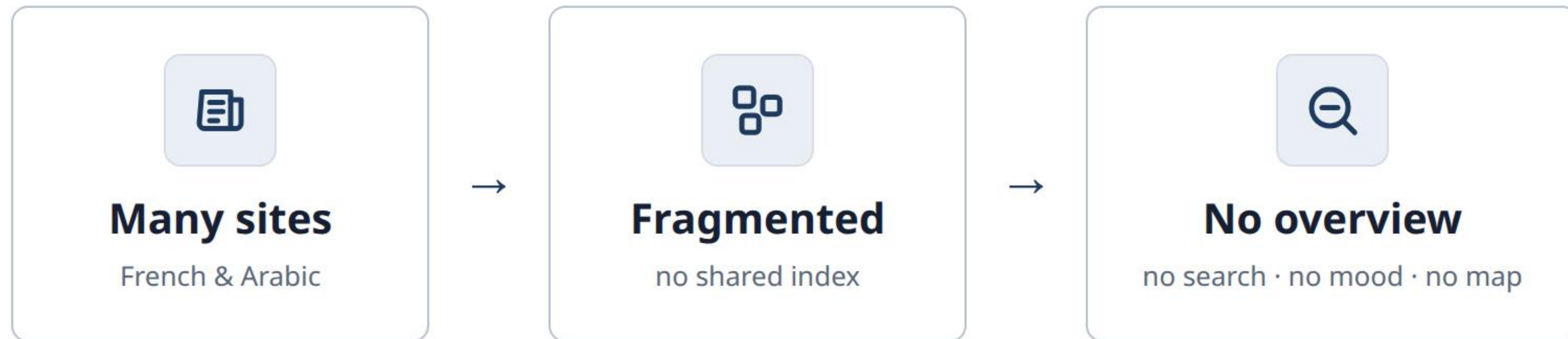


News Insight

One place to read and understand Tunisian news



Tunisian news is everywhere, and impossible to see as a whole.



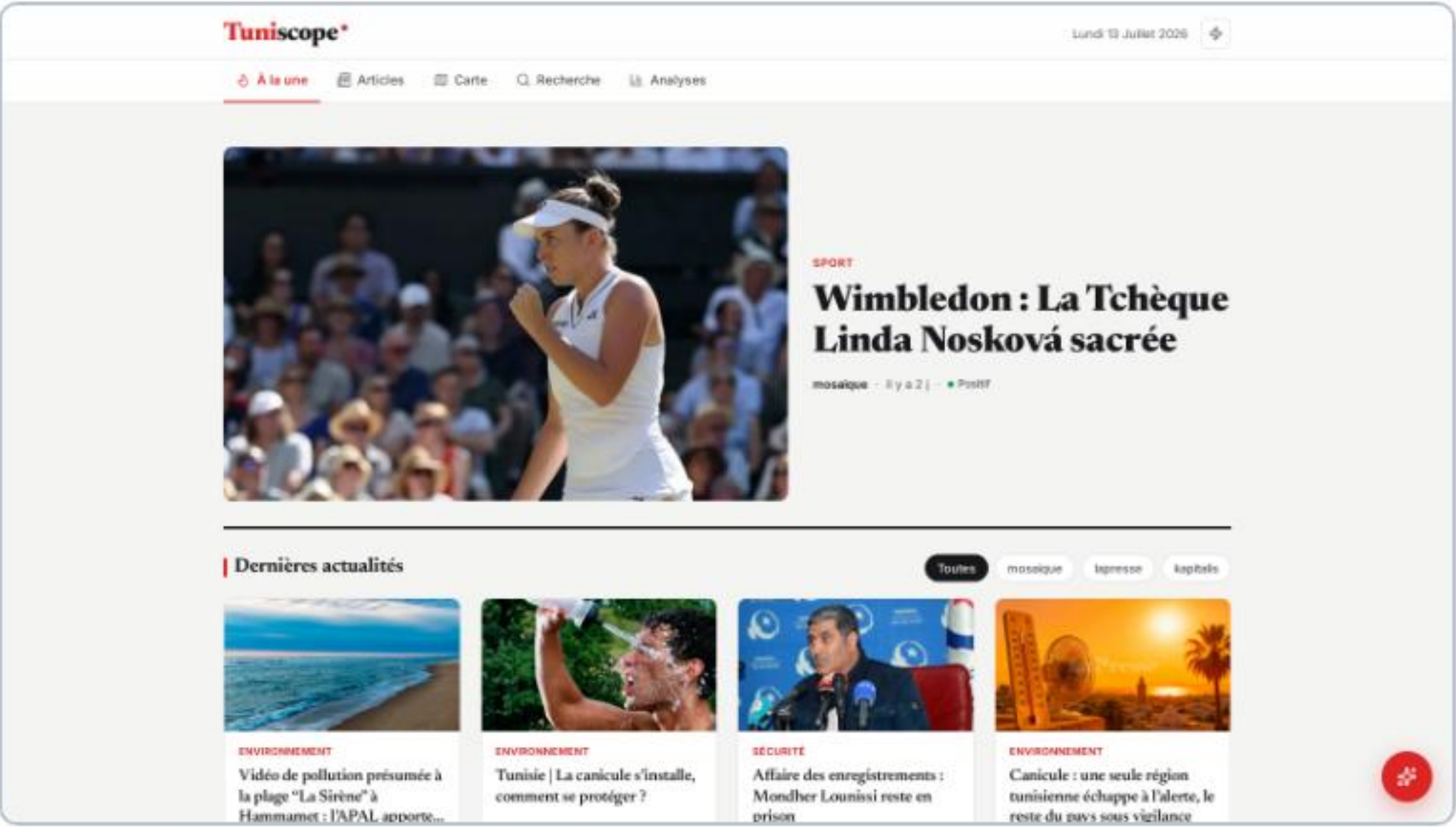
The gap: you check each site by hand, and still can't search by meaning, gauge the overall mood, or see which regions are in the news.

One pipeline: collect the news, understand it, explore it.

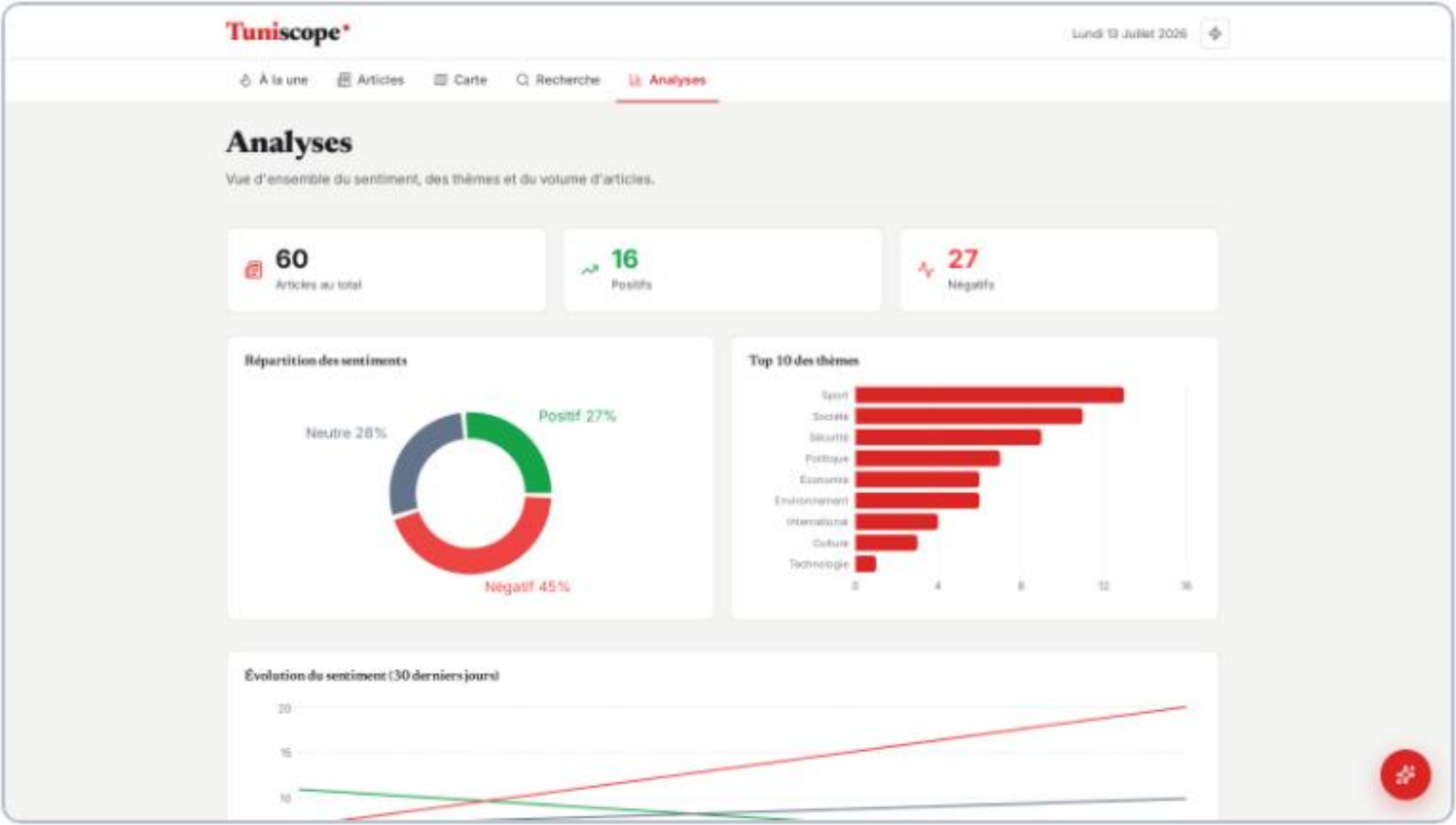


It runs on its own, and everything reaches you through one web app.

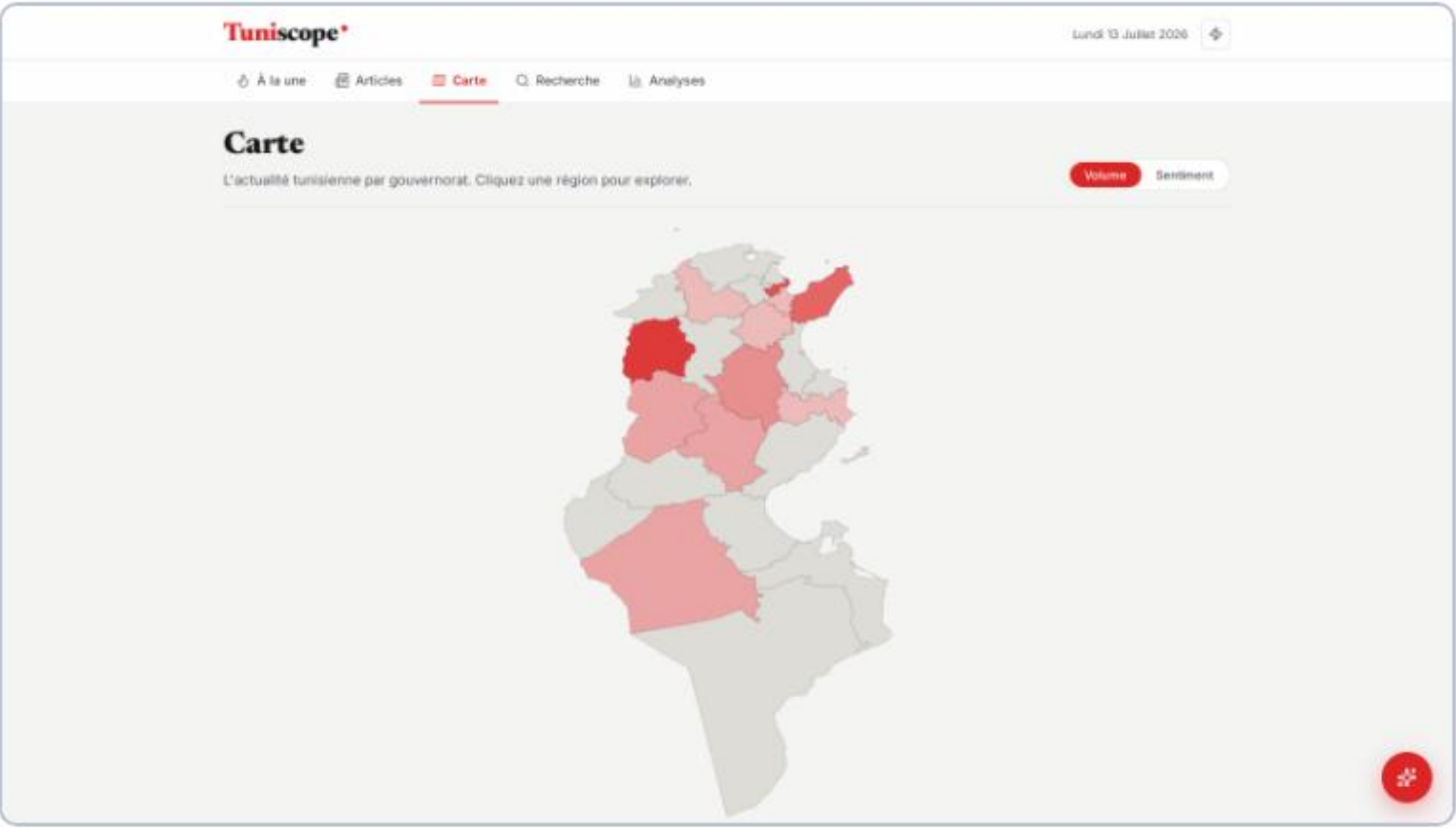
Read, analyse, map and ask, in one app.



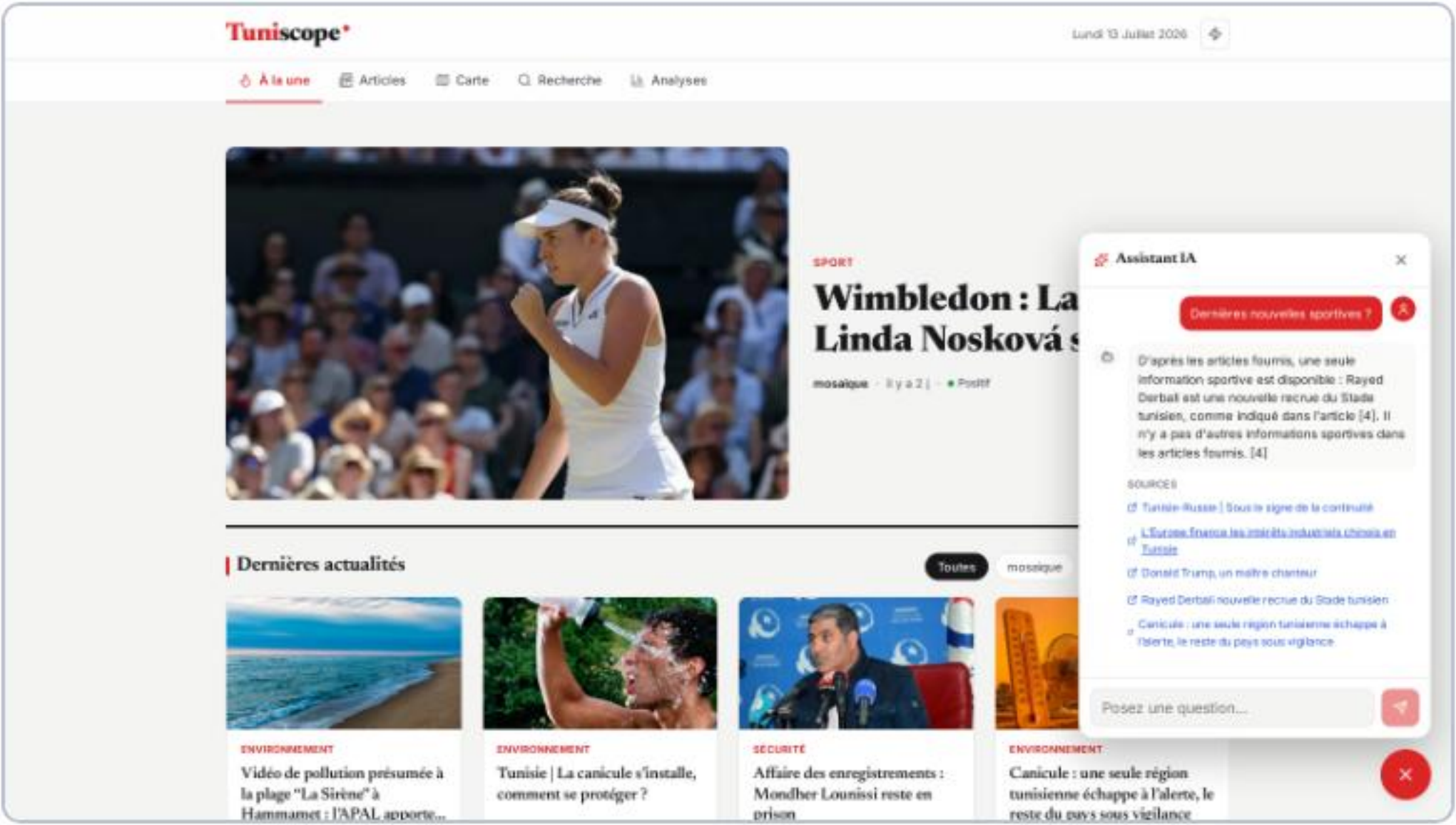
Read · every source in one feed



Analyse · sentiment, topics, trends



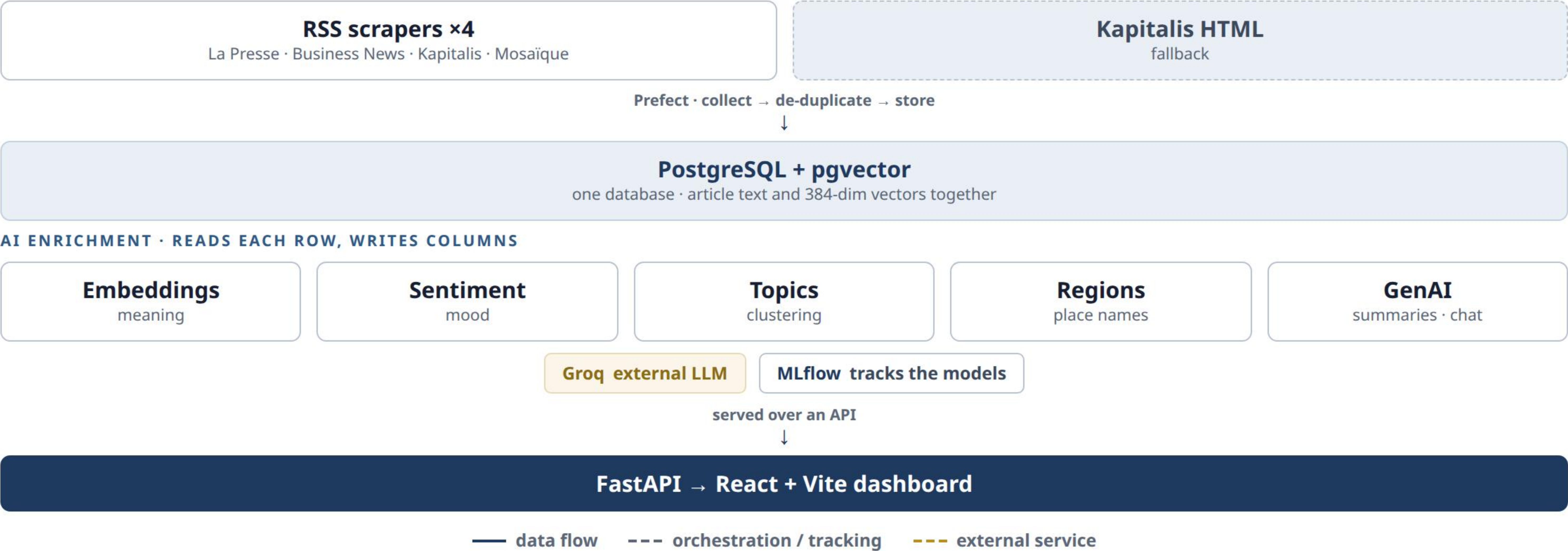
Map · news across the 24 governorates



Ask · an AI assistant that cites sources

The whole system, from sources to dashboard.

SOURCES



Every article is read by AI and turned into data.

Meaning

The text becomes a numeric fingerprint (an embedding), so similar articles sit close together in vector space.

multilingual-e5 · vector(384)

Mood

A language model labels the tone of each article.

positive

neutral

negative

Topic

Articles are grouped into recurring topics automatically, with no preset list, then each cluster is named.

BERTopic clustering

Region

Place names in the text are matched to one of Tunisia's 24 governorates.

gazetteer · Sfax · Tunis...

Why it matters: free-form text becomes structured signals you can search, chart and map.

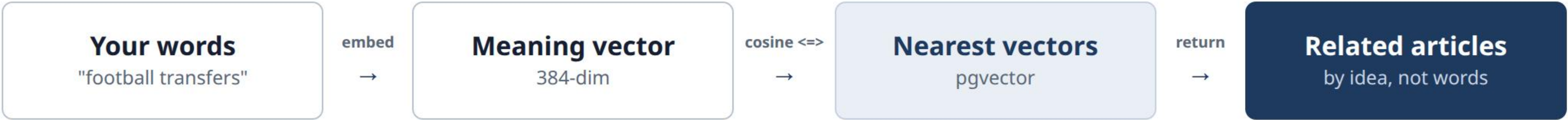
One database holds the text and its meaning together.

Each article and its 384-dim vector live in the same PostgreSQL row, so "find similar articles" is a single SQL query with `pgvector`, not a second system to run.

	pgvector (our choice)	A dedicated vector database
Infrastructure	None extra: it is just Postgres	A second system to run and keep in sync
Consistency	One transaction for data and vectors	Two stores that can drift apart
Scale ceiling	Millions of rows (with an index)	Billions, very high query rates
Best for	This project's size	Massive, high-traffic retrieval

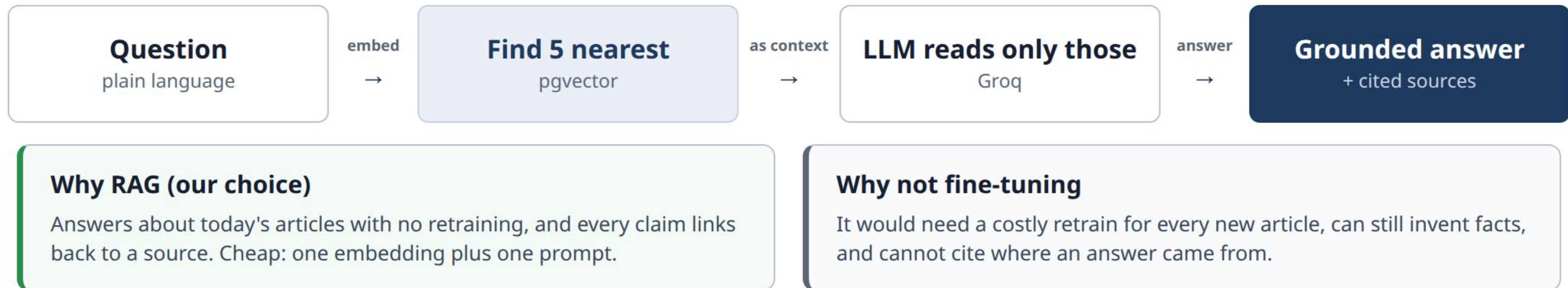
The trade-off: at this corpus size, one database beats running two and syncing them. We would switch only if the corpus grew by orders of magnitude.

Search by meaning, not just keywords.



Why it beats keyword search: it finds articles about clubs, matches and signings even when they never contain the word you typed, and it works across French and Arabic at once.

The assistant answers from real articles, not from memory.



It runs itself, records what it does, and rebuilds with one command.

Orchestration

A Prefect flow runs collect → transform → load on a daily schedule or on demand, with automatic retries.

Prefect · idempotent (upsert)

Tracking

Every clustering run's settings, metrics and fitted model are recorded, so any run can be reproduced.

MLflow · params · metrics · model

Monitoring

Each run appends one row: articles ingested, topics found, outlier rate, sentiment split.

pipeline_metrics

Reproducibility

The whole stack (database, API, app) starts together on any machine with Docker.

docker-compose up

Why it matters: orchestration, tracking, monitoring and reproducibility are what make this an operated platform, not a one-off notebook.

Every tool was chosen for fit, not fashion.

Layer	Tool	Why this one
Data & ML	Python	The richest ecosystem for scraping, NLP and ML
Storage + search	PostgreSQL + pgvector	One store for article data and its vectors
Language models	Hugging Face (e5, XLM-R)	Multilingual and CPU-friendly, for FR + AR text
Topics	BERTopic	Discovers topics with no preset list of labels
Orchestration	Prefect	Scheduling, retries and safe re-runs
Tracking	MLflow	Makes stochastic clustering reproducible
Backend / frontend	FastAPI · React + Vite	A typed API and a fast, chart-rich dashboard
Deployment	Docker	One-command, reproducible setup anywhere

One place to read and understand Tunisian news.

1 **Search by meaning, not just keywords**

Find related articles even when they use different words, across French and Arabic.

2 **An AI assistant that answers with sources**

Ask a question in plain language and get an answer grounded in real articles.

3 **Mood, topics and regions at a glance**

A dashboard and a map turn the whole feed into one clear picture.

An automated, tracked and reproducible pipeline, built with **Python · PostgreSQL + pgvector · Prefect · MLflow · FastAPI · React · Docker**.

Thank you

News Insight · Questions & discussion